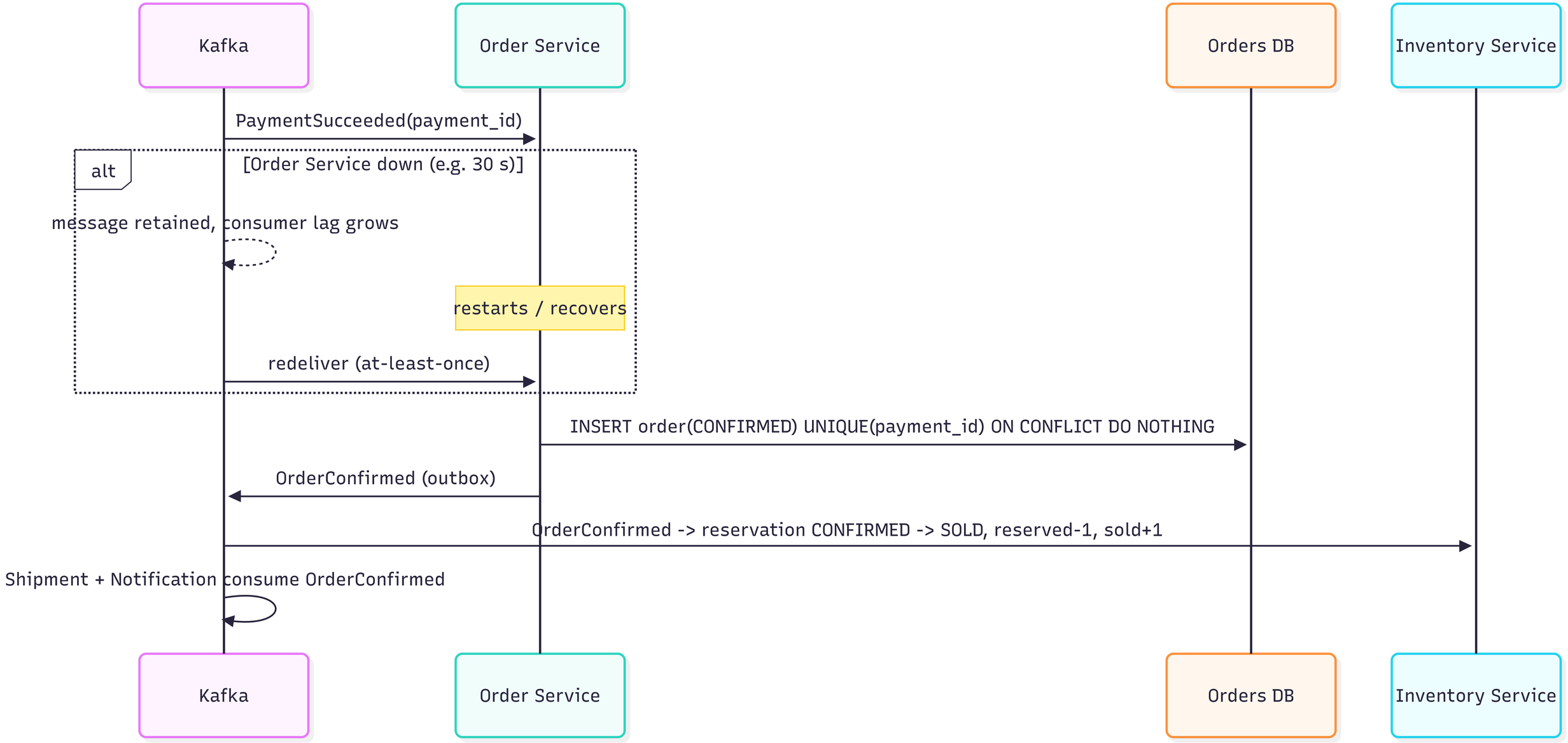

Kafka
Order Service
Orders DB
Inventory Service
PaymentSucceeded(payment_id)
alt
[Order Service down (e.g. 30 s)]
message retained, consumer lag grows
restarts / recovers
redeliver (at-least-once)
INSERT order(CONFIRMED) UNIQUE(payment_id) ON CONFLICT DO NOTHING
OrderConfirmed (outbox)
OrderConfirmed -> reservation CONFIRMED -> SOLD, reserved-1, sold+1
Shipment + Notification consume OrderConfirmed
Kafka
Order Service
Orders DB
Inventory Service